

Cake Cutting

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1 Piece of cake? How many do you expect?

You must have heard the expression “piece of cake” many times. Have you thought of how many pieces should be expected after making n random cuts? This probability question is easier asked than solved. The answer is $\frac{(n+2)(n+3)}{6}$, which is easy to verify for no cuts, since the cake will be in the original one piece. For one cut, the formula produces 2 pieces: $\frac{(1+2)(1+3)}{6} = 3 \cdot 46 = 2$. The strict probability theory formulation is to produce “the expected number of pieces when a round circular cake is cut with n random cuts”. Let’s count cut-by-cut!

2 Proof

2.1 Cuts do not intersect

First, it is easy to see that if we require for cuts not to intersect, the n th cut will cut one of the existing whole pieces into two new ones, increasing the overall count by 1. Since we started with the whole cake as one piece, after n cuts this will result in $n + 1$ pieces. Therefore, $n + 1$ is a minimum number of pieces from cutting.

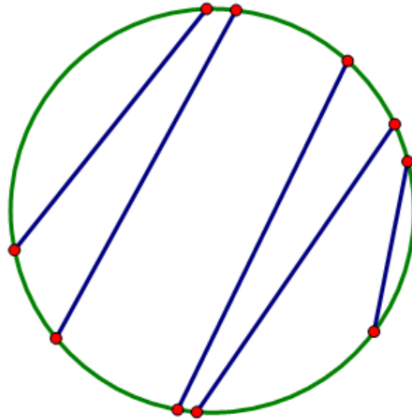


Figure 1: n cuts, no intersections.

2.2 Adding intersections

Second, let's see what is the expected value of pieces added due to an intersection of cuts. Each intersection adds a piece to the total count. If $n = 2$ and the second cut is truly random, what is the expected value of cuts added to the established minimum? To count, please notice that a random cut is equivalent to randomly choosing two points on the circle, which is the circumference of the cake. Then the line through these two points is the random cut.

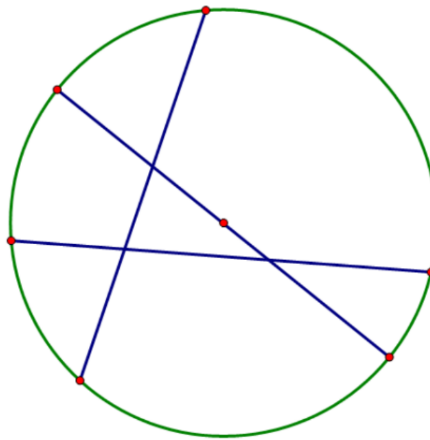


Figure 2: Three cuts, three intersections.

2.3 Case of two cuts intersecting

Two random cuts will be equivalent to picking four random points on the circle. Let's assume that the four points were chosen but the decision about pairing them into straight lines cuts is unknown. Also, without loss of generality, let's pick one of these points, denoting it with A , and other points clockwise as B , C , and D . Then, the probability of $\overline{AB}$ being the cut is $\frac{1}{3}$ (chosen between $\overline{AB}$, $\overline{AC}$, and $\overline{AD}$), and automatically leads to the total of three pieces. This is because the other cut has to be $\overline{CD}$ and they don't intersect.

The choice of $\overline{AC}$ has a $\frac{1}{3}$ probability of being the cut, and results in the total of four pieces, since the other cut has to be $\overline{BD}$ necessarily producing an intersection and one additional, fourth, piece of cake.

The choice of $\overline{AD}$ is symmetric to the choice $\overline{AB}$, has $\frac{1}{3}$ probability of being the cut and results in three pieces total.

The above proves that the expected value of pieces added to the established minimum due to an intersection of two random cuts is $\frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 0 = \frac{1}{3}$.

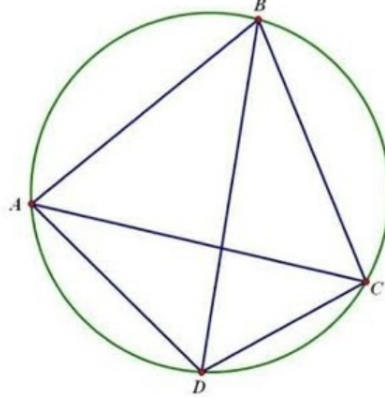


Figure 3: Four points with all possible cuts through them.

3 Generalizations

3.1 N cuts intersecting

With n cuts, let's focus on a certain cut. It could produce an intersection with any of the other cuts, as they are all random, and each pair of cuts is random, similar to the second step above for $n = 2$. There are $(n - 1)$ cut pairs to consider with the cut in focus. Therefore, per the result in 2.3, the expected number of pieces added due to intersections with the cut in focus is (*expected value of pieces added per each pair*)(*#pairs*) = $\frac{1}{3} \cdot (n - 1) = \frac{n-1}{3}$. However, due to symmetry, this logic applies to each of the n cuts. Applying this logic to every cut of n will double-count each intersection. Therefore, the expected value of additional pieces due to pairwise intersections within n cuts will be

$$E(\#pieces\ added\ due\ to\ cuts) = \frac{n}{2} \cdot \frac{n-1}{3} = \frac{n \cdot (n-1)}{6}.$$

4 Conclusion

Bringing in the first step that without intersections there are $n + 1$ pieces,

$$E(\#pieces\ from\ n\ random\ cuts) = n + 1 + E(\#pieces\ from\ intersections) = n + 1 + \frac{n \cdot (n-1)}{6} = \frac{6 \cdot n + 6 + n^2 - n}{6} = \frac{n^2 + 5 \cdot n + 6}{6} = \frac{(n+2) \cdot (n+3)}{6}.$$